

EMMANUEL DUODU

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Data Systems & Analytics engineer with 5+ years of experience driving measurable business impact through ML, experimentation, and product analytics. Built A/B testing frameworks, churn models, and personalized pricing experiments that lifted conversion 18% and cut churn 14% across users.

WORK EXPERIENCE

Quant Technologies - Remote

06/2024 - 05/2025

Data Analytics engineer

- Segmented 2M+ customers using K-Means and RFM analysis to identify 5 personas, enabling a targeted email campaign that lifted CTR 21% versus the prior generic blast.
- Reduced new-user churn by 14% within 3 months by developing an XGBoost model that identified at-risk users 7 days earlier, enabling timely retention strategies.
- Scaled the experimentation platform from 20 to 120+ tests per quarter by designing a self-serve A/B testing framework in Python, cutting analyst review time 65% and onboarding 40+ stakeholders.
- Built predictive churn models using logistic regression and XGBoost to prioritize retention follow-ups.

YBI Foundation

09/2021 - 11/2023

Data Scientist(Intern)

- Conduct time-series forecasting in IBM SPSS to analyze network usage trends, supporting a 17% reduction in peak-time congestion.
- Analyzed transaction datasets in Microsoft SQL Server to identify fraud patterns, contributing to 29% reduction in fraudulent activity.

EDUCATION

Master of Business Administration, Information Technology

Academy Europe Open University

01/2022 - 04/2025

Relevant Coursework – Statistics, SQL, Big Data Analytics, Predictive Modelling, Data Mining, and Machine Learning.

Professional Certificate, Database and Software Engineering

IPMC University College of Technology • Ghana

07/2018 - 08/2021

PROJECTS

AI Data Analyst with Claude Code

Built an agentic analytics system with 25 subagents and 45 skills in markdown, turning Claude Code into a full analytics team that connects to data warehouses and answers business questions in plain English. Cut ad-hoc analysis time from hours to minutes.

SKILLS

Core: PySpark, Python (Pandas, Scikit-Learn, Statsmodels), R, SQL, Oracle, Databricks

Machine Learning: Feature Engineering, Model Evaluation (Precision, Recall, AUC-ROC), Supervised Learning (Logistic Regression, Random Forest, XGBoost, Gradient Boosting), Unsupervised Learning (K-Means, Hierarchical Clustering)

Experimentation & Statistics: A/B Testing, Hypothesis Testing, Multivariate Testing, Power Analysis

Product & Risk Analytics: Behavioral Risk Signals, Cohort Analysis, Conversion Optimization, Funnel Analysis, Pattern Recognition, Retention Analysis, Risk Modeling, User Behavior Analysis, User Segmentation

Cloud & Data Engineering: AWS (S3, Glue, Athena, SageMaker), Azure, Databricks, dbt, Snowflake

Visualization: Looker, Power BI, Tableau, Excel, Google Sheets